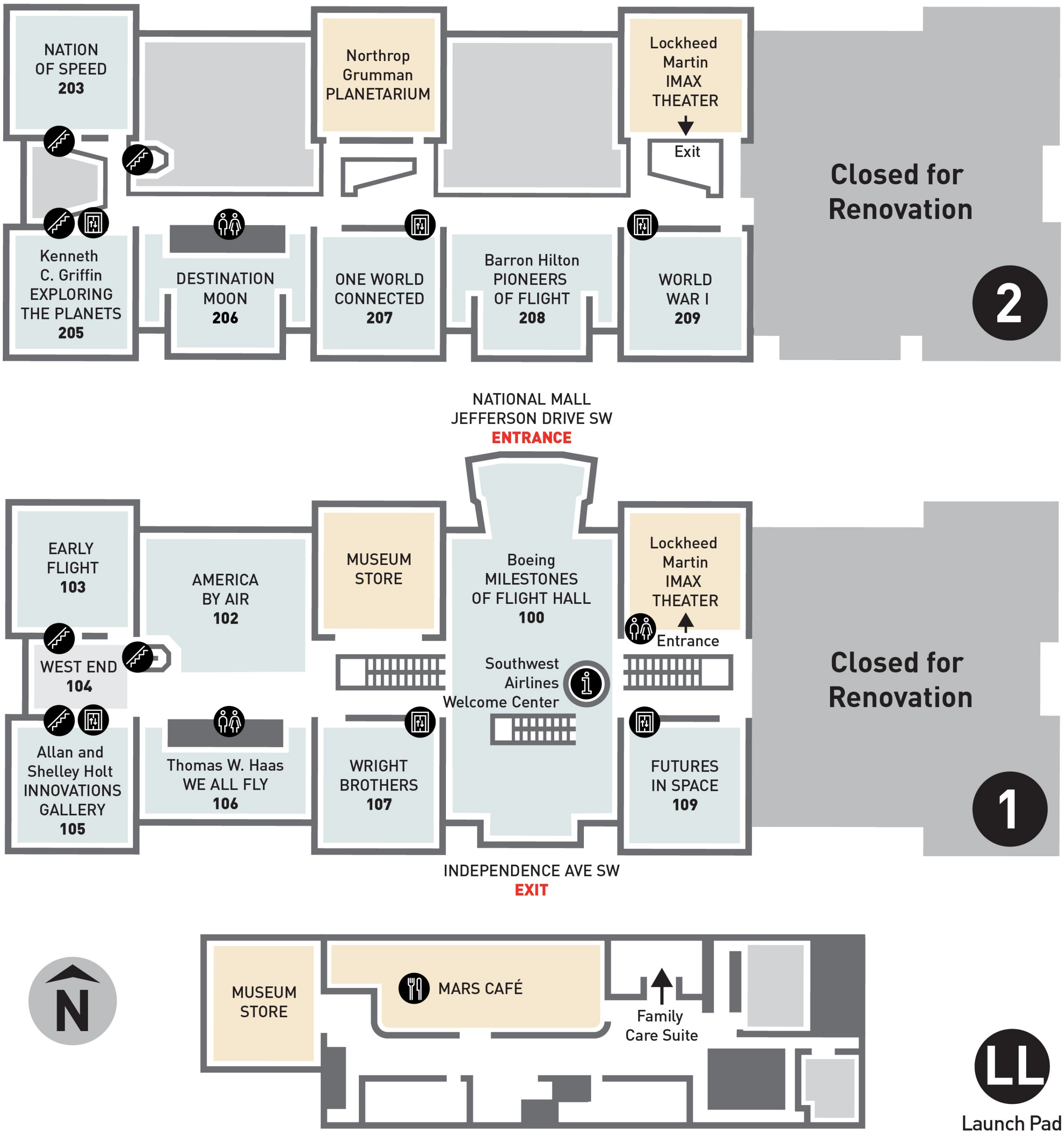
Smithsonian National Air and Space Museum
Self-guided tour, The Planetary Society
April 19, 2026



This tour concentrates on five galleries in the Mall museum: Milestones of Flight, Destination Moon, Exploring the Planets, One World Connected, and Futures in Space. Of these, the first three galleries are likely the favorites. This tour also notes some artifacts in the Commons (areas outside galleries).

Three additional galleries particularly of interest will open later this year: Discovering Our Universe (the new astronomy gallery) and Living in the Space Age (which includes Skylab, the Hubble Space Telescope, rockets in the Missile Pit, and more) will open on July 1, 2026, and At Home in Space (featuring the ISS) will open on October 30, 2026.

Please note that the ONLY entrance to the museum is from the Mall side, at the midpoint of the museum. The winged portico on the Mall has doors on both its East (4th Street) and West-facing (7th Street) sides; the East line for entry is typically shorter. You can exit through both the Mall and Independence Avenue doors. Once you've been scanned into the museum, you can stay until closing. If you need to leave and come back, your ticket allows one readmission. Upon your return, don't get into the line for the next hour's timed entry; come straight up to the doors and explain you're returning.

<u>Milestones of Flight Gallery</u>

As you enter the museum through this gallery, look at the floor: it displays the **pulsar map** launched on Pioneers 10 and 11 and on both Voyager spacecraft, with the frequencies adjusted to mark Earth in July 2026, celebrating the 50th anniversary of the opening of the original museum.

Major space artifacts on the ground on the right-hand side of the gallery include:

- **Don't miss the touchable Moon rock brought back by Apollo 17! It's in the curved silver display by the LM leg closest to the escalator.**
- **Viking lander:** Proof test article for Viking 1 (launched 8/20/1975; landed 7/20/1976; last transmission 11/11/1982) and Viking 2 (launched 9/9/1975; landed 9/3/1976; last transmission 4/11/1980). Viking 1, en route to Mars, sent the radio signal that triggered the ribbon cutting opening the Museum on 7/1/1976.
- **Mercury capsule** *Friendship 7:* John Glenn became first American in orbit on 2/20/1962.
- **Gemini IV capsule:** Jim McDivitt and Ed White, 6/3-7/1965, first American spacewalk.
- **Apollo Lunar Module:** LM-2, built for an Earth-orbit test flight deemed unnecessary when the first test flown on Apollo 5 was successful, was used instead for ground testing; restored to look exactly like LM-5, the *Eagle* of Apollo 11, the day it landed on the Moon, 7/20/1969.

Space artifacts hanging on the right-hand side of the gallery include:

- ***SpaceShipOne*****:** First non-government crewed vehicle to reach space twice within two weeks, winning Ansari X-Prize in 2004.
- **Sputnik 1:** Replica of first satellite ever launched, 10/4/1957.

- **Explorer I:** Backup for first American satellite launched 1/31/1958; final transmission 5/23/1958; burned up on reentry 3/31/1970. Onboard instruments from Dr. James Van Allen led to discovery of Van Allen Belts, charged particles trapped by Earth's magnetic field.
- **Mariner 2:** Model of Venus probe constructed by JPL from test components. Probe launched 8/27/1962; radioed data from Venus 12/14/1962. First to measure solar wind. Contact lost 1/02/1963; still in solar orbit.
- **Pioneer 10:** Full-scale model from spare parts. Original launched 3/2/1972, entered Asteroid Belt 7/15/1972, Jupiter fly-by 12/3/1973, crossed Neptune's orbit June 1983. Last telemetry received 4/27/2002; signal dropped below threshold for detection February 2003.
- **Lunar Orbiter:** Engineering mock-up by Boeing. Five Lunar Orbiters were launched to map about 99% of the Moon's surface in 1966-1967, used to select Apollo landing sites. When all their film was depleted, all five were crashed deliberately into the Moon.

Fans of science fiction should walk toward the Independence Avenue exit. On the east side of the museum's South Lobby, at the top of the escalator/stair down to the lower level, you will find the **Starship *Enterprise***, the studio model used throughout the original *Star Trek* series in the 1960's. The model lights up every hour on the hour, for only five minutes; don't miss it! The camera only ever saw the model from one side; walk around it toward the Robert McCall mural on the wall, and you'll realize the far side of the model is unfinished, and has a hole in it so the wires powering the internal light bulbs could be plugged in! (Why does NASM display a fictional starship? Because everything begins with the imagination – and dreams inspire careers in discovery ...)

Head upstairs to the second floor using the escalator near the Lunar Module, or the elevator in the entry to the Wright Brothers' gallery. (That elevator's rear wall is the view from inside the Lunar Module!)

Above the West side escalators is **Poe Dameron's X-Wing** from *Star Wars: The Rise of Skywalker*. It's the only artifact in the museum with two distinct signs, located on the railing between the escalators and the walkway to the 747 nose. One is written in the real world ("This is a movie prop") while the other is written *inside* the world of *Star Wars,* as if it's all real ("T-70 Resistance fighter! Armaments! Engines! Poe Dameron flying it!"). (NASM curators are the most marvelous nerds on the planet, and we love them dearly.)

Three spacecraft hang in the West Commons area outside the galleries. They are:

- **ATS-1:** Launched in 1966, Applications Technology Satellite 1 was the first of six satellites built by Hughes for NASA to test new technologies in space communications and observation from geosynchronous orbit. Its 3-year mission lasted for 19 years, until 1985. Our ATS-1 is a prototype built by Hughes.
- **Clementine:** Built by the Naval Research Laboratory and launched in January 1994, it mapped the entire surface of the Moon, transmitting 1.8 million images. Its radar data suggested the existence of ice in a crater at the South Pole,

confirmed by NASA's Lunar Prospector in 1998. This is an NRL engineering model.
- **Mariner 5:** Launched to Venus 6/14/1967 to obtain data on the composition of its atmosphere and its radiation and magnetic field environment, its October 1967 flyby provided an understanding of the extremely hot surface temperatures and unexpectedly dense atmosphere. Its data were used in conjunction with findings from the USSR's Venera 4 lander. This is a full scale model constructed by JPL from spare parts.

<u>Destination Moon Gallery</u>

If the development of human spaceflight interests you, this is your gallery. It has two levels; do not miss the second floor, which includes exhibits on spacesuit design and construction, the role of Mission Control (featuring the story of Apollo 13), introduces the **Ranger** and **Surveyor** landers that helped inform Apollo operations, and displays an actual **F-1 engine from a Saturn V**. Make sure you look up while on the first level of the gallery, because you will walk right under that 18,000-pound rocket engine! (Make certain you look up *wherever* you walk in the museum; NASM's artifacts fly, and if you're not looking up, you'll miss half the museum!)

Do take time for the video displays in this gallery, which include JFK's Rice University speech in the first part of the gallery (beside **Alan Shepard's spacesuit** and **Mercury capsule *Freedom 7***), and a stunning History Channel timeline in the center portion that puts the Space Race into the context of world events and specifically into contemporaneous developments in American society and culture.

Other key artifacts in the gallery include:

- **Gemini VII capsule:** The 14-day mission of Jim Lovell and Frank Borman in 1965.
- **Neil Armstrong's Moonwalk spacesuit:** Note that the suit's boot does not match the famous boot prints on the Moon; the astronauts wore insulated overshoes to protect their suit boots from the 250˚ F heat of sunlit ground, and most left them on the Moon to save weight. **Gene Cernan's overshoes** are the last exhibit on the left as you exit the gallery.
- **Apollo 11 Command Module *Columbia*:** Walk around behind the capsule for a sobering look at the damage to its ablative heat shield from the heat of reentry. The CM's hatch is in a separate case that explains changes made after the Apollo 1 fire on the pad.
- **Apollo 11 F-1 engine parts:** The glass case west of the suspended F-1 engine contains pieces of one of the first-stage engines from the Saturn V that launched Apollo 11, retrieved from the bottom of the Atlantic by a Bezos-funded expedition in 2013.
- **Lunar Roving Vehicle:** Apollo 15, 16, and 17 included battery-powered “Moon buggies” built by Boeing and General Motors that allowed the astronauts to travel as far as 4.7 miles from the LM. This one was a “qualification test unit,” subjected to more severe conditions than expected on any actual mission.
- **Lunar Reconnaissance Orbiter:** Built by the Goddard Space Flight Center and

launched in 2009, the LRO entered an eccentric polar mapping orbit (12 mi by 103 mi) on 9/15/2009, and has fuel to continue operating until at least 2027. It has mapped over 98% of the lunar surface at 330-ft resolution, with better than 2-ft resolution of all Apollo landing sites. This is a full-scale model used in structural verification tests. (Look up near the exit from the gallery.)

Exploring the Planets Gallery

When you enter this gallery, you're coming into our solar system from outside, taking a tour from the outer icy worlds to the inner rocky ones. Be certain to look at the floor as well as up at everything suspended from the ceiling and described on the walls; all dimensions are fully in play. Interactive consoles let you explore worlds and topics as you tour.

At the center of the gallery is a theater in the round featuring a 7-minute, silent video called *Walking on Other Worlds,* which uses photographs and extrapolations from data from missions that landed on other bodies in our solar system to convey you to those target planets, moons, and asteroids, and then give you a 360-degree view of your landing site. On approach to each landing, you can look at either hemispherical screen for data on the mission; as soon as you land, be sure to look all the way around!

Key artifacts in this gallery include:

- **Voyager Record Cover (Duplicate):** Ahead and to the right when you enter, you'll see a picture of Carl Sagan near the case containing the Voyager Golden Record. The diagrams on the cover include the pulsar map identifying Earth as the origin, as well as diagrams to assist any beings who might find either of the Voyager spacecraft to access and interpret the data on the record.
- **Stardust Comet Sample Return Capsule:** Built by Lockheed Martin and launched on 2/7/1999, it collected dust and ice samples from comet Wild 2 in 2004, and parachuted to Earth on 1/15/2006. It was the first US vehicle to return extraterrestrial material from beyond the Moon. The collector grid is shown deployed for flight and rotated 180 degrees for display. Stardust revealed that most of a comet's rocky matter formed inside our solar system rather than beyond the orbit of Neptune, as previously believed.
- **Voyager:** Hanging high near the rear of the gallery, directly ahead of you as you enter, is the Development Test Model of the Voyager 1 (launched 9/5/1977) and 2 (launched 8/20/1977) spacecraft, manufactured by JPL from facsimile and dummy parts.
- **Mars Rovers:** One case contains three generations of US Mars rovers.
  - The ***Marie Curie*** was the flight spare for our first rover, *Sojourner*, which launched 12/4/1996 and landed 7/4/1997. It operated for 83 sols (Martian days), until communications with its lander were lost on 9/27/1997. JPL operated *Marie Curie* on Earth to test commands before sending them to *Sojourner* on Mars.
  - The ***Mars Exploration Rover Surface System Test Bed (MER SSTB)*** was the Earth-based control for the two MER rovers, *Spirit* (launched 6/2003; end of mission 5/2011) and Opportunity (launched 7/2003; end of mission 2/2019).

JPL used the MER SSTB to test rover operations and troubleshoot problems.

  - The **Mars Science Laboratory (MSL) *Curiosity*** launched in 11/2011, landed in Gale Crater in 8/2012, and is still chugging along long after its one Martian year (687 days) scheduled mission. NASM's craft is a medium-fidelity, full-scale model on loan from JPL. (NASM does not have an analogue on display for the Mars 2020 *Perseverance*, which launched 7/30/2020 and landed on 2/18/2021. It is similar to but is larger and about 275 pounds heavier than *Curiosity*. NASM does have the full-scale prototype of the *Ingenuity* Mars helicopter on display at the Steven F. Udvar-Hazy Center.)
- **Mariner 10:** The first spacecraft to Mercury, it launched 11/3/1973, reached Venus on 2/5/1974, and became the first to use gravity assist to change its flight path, crossing the orbit of Mercury three times, on 3/29/1974, 9/21/1974, and 3/16/1975. This is the flight spare.
- **Spock's Ears and a Tribble:** *Star Trek* is represented here by a set of Vulcan ear tips worn by Leonard Nimoy and a Tribble.
- **Surveyor III Camera:** In 11/1969, the crew of Apollo 12 – Pete Conrad and Alan Bean – landed on the Moon with precision, walking distance away from the landing site of Surveyor III, and brought its TV camera and a few other instruments back to Earth.

One World Connected Gallery

This second-floor gallery focuses on how aviation and space – especially satellites – have totally changed the way we live on Earth: how we travel, how we communicate, how we farm, how we understand weather and climate, how we study our world from high vantage. People interested in communications technology and Earth observation satellites may particularly enjoy visiting this gallery. The floor is an integral part of the display, as it defines the different orbital locations – low, medium, high – relative to the planet. The center of the gallery is an interactive built around a large Earth globe, where you can call up depictions of population concentration, transportation, and wildlife tracking. There's also a mockup of the cupola from the International Space Station, where you'll hear the constant ambient noise of the station's fans and equipment.

The interactives in this gallery let you explore all these things, and contribute your answers to questions other visitors have also addressed. Most of the major artifacts in this gallery are representative of key communications and observation satellites, including **Intelsat II, Iridium, Relay 1, Sirius FM-4, TIROS 1 (Television Infrared Observation Satellite,** first weather satellite, 4/1/1960), and **ATS 6**.

Futures in Space Gallery

This new first-floor gallery includes current activities in space and asks visitors to engage with questions. Why go to space? Who gets to go, and what would they do there? How would humans live in space or on other worlds? What roles might robots play?

Commercial space travel didn't exist when the museum first opened. Now, in addition to scientific exploration, there's space tourism. Launch has gone from being exclusively the province of governments to a competitive commercial venture, including offering small satellite “ridesharing” that provides low cost launch services even to small businesses or educational institutions. Many countries are participating in space. Private companies are building exploration and landing vehicles not just for government contracts, but for their own purposes. This gallery asks guests to share their ideas.

Artifacts in this gallery include:

- ***Star Wars* R2-D2 Model:** Built by Adam Savage, this is part of imagining robots.
- **Blue Origin New Shepard Crew Capsule Mock-Up:** This is standing in for the capsule Blue Origin is currently using to launch suborbital space tourists; when the *RSS First Step* is retired from launch, it will replace the mock-up.
- **SpaceX Falcon 9 Merlin 1D Engine and Grid Fin:** Reusable rocket boosters have changed the launch universe. This engine was used in three launches. The grid fin enables steering a booster back to landing. Both were donated by Space X in 2022.
- **Rutherford Engine, Rocket Lab:** New technologies in space include rocket engines with 3-D printed parts, allowing for fast production. The Electron rocket using this engine launches small satellites.
- **Virgin Galactic Rocket Motor Two:7** This hybrid engine, flown on *SpaceShipTwo*, launched a suborbital flight with three crew and three paying tourists on 6/29/2023. It combines a solid propellant with the consistency of rubber with liquid oxygen as its oxidizer, allowing the engine to produce variable thrust and to be turned off and restarted in flight; something not possible for a solid rocket booster.